# ADCS TELEMETRY ANOMALY CATALOGUE

ESA OPS-SAT ? Complete Fault Classification and Explanation Guide
Version 1.0 | Anomaly Reference Manual

## SECTION A: ANOMALY TAXONOMY

Satellite telemetry anomalies are classified into four primary types:

TYPE 1 ? POINT ANOMALY:
  A single data point deviates significantly from the expected distribution.
  Example: A single magnetometer reading spikes to 120 ?T before returning
  to normal. Often caused by cosmic ray bit flips in ADC registers, brief
  electromagnetic pulses, or single-event upsets (SEU) in the sensor circuit.
  Severity: LOW to MEDIUM. Typically self-clearing.

TYPE 2 ? CONTEXTUAL ANOMALY:
  A data point appears normal in isolation but is anomalous in its temporal
  or mission context. Example: A magnetometer reading of 45 ?T is within
  the nominal range, but it is constant for 10 minutes when the orbital
  dynamics predict it should be varying smoothly ? this indicates a sensor
  frozen on a cached or stuck value.
  Severity: MEDIUM to HIGH. May require operator intervention.

TYPE 3 ? COLLECTIVE ANOMALY:
  A sequence of data points that is individually plausible but abnormal as
  a pattern. Example: All three magnetometer channels simultaneously showing
  elevated noise variance over several orbital passes, suggesting a system-
  level issue (EMI source turned on, grounding problem, power supply noise).
  Severity: MEDIUM to HIGH. Requires subsystem-level investigation.

TYPE 4 ? DRIFT ANOMALY:
  A slow, gradual change from nominal baseline over days or weeks. Example:
  A photodiode channel showing progressively decreasing peak voltage over
  several weeks, indicating solar cell degradation or sensor degradation.
  Severity: LOW initially, escalating. Requires trend monitoring.

## SECTION B: MAGNETOMETER ANOMALY CATALOGUE

### B.1 ? SENSOR BIAS DRIFT

Channels Affected: CADC0872, CADC0873, and/or CADC0874
Anomaly Type: Type 2 (Contextual) or Type 4 (Drift)

DESCRIPTION:
  A constant offset (bias) is added to the magnetometer output, causing
  systematic deviation from the true Earth magnetic field vector. The signal
  shape and temporal variation are preserved but shifted by a fixed amount.
  The reading is not random noise ? it is a clean offset.

SYMPTOM SIGNATURE IN TELEMETRY:
  - The affected channel shows a consistent +/- offset compared to the
    expected IGRF model prediction for the current orbital position
  - The bias is stable over multiple orbits (Type 4) or appears suddenly
    (Type 2)
  - Nominal autocorrelation at lag-1 (?? > 0.95), but mean (?) shifted

  - The signal variance is not elevated ? only the mean is wrong

ROOT CAUSES:
  1. Magnetic contamination from nearby electronics or harnesses
     carrying current (e.g., EPS switching noise, battery charge current)
  2. Residual magnetisation of satellite structural components
  3. Magnetorquer residual dipole after power-off (slow decay: minutes)
  4. Temperature-induced bias change in the sensor electronics
  5. Radiation damage causing permanent offset in sensor circuitry
  6. Improper factory calibration constants stored in OBC memory

DIAGNOSTIC CHECKS:
  - Compare all three axes: if only one axis is affected, likely a local
    interference source near that sensor axis
  - Check magnetorquer state: if magnetorquers were recently active, the
    bias may be a residual magnetic moment (expected to decay in 5-15 min)
  - Cross-reference with power subsystem telemetry: EPS switching events
    often correlate with sudden magnetometer bias changes
  - Check if bias persists across orbital phase changes: if it changes with
    sunlit/eclipse transitions, temperature is likely the cause
  - Review recent software uploads or calibration table modifications

SEVERITY: MEDIUM ? attitude estimation error increases proportionally with
bias magnitude. For attitude determination accuracy, a 5 ?T bias causes
approximately 3-5 degrees of pointing error at LEO.

CORRECTIVE ACTIONS (in priority order):
  1. IMMEDIATE: Note orbital position and time. Record bias value for all
     three axes. Compare with IGRF model prediction for current position.
  2. Wait 15 minutes: If bias appeared immediately after magnetorquer
     activity, it may self-correct as residual magnetic field decays.
  3. Command magnetometer recalibration procedure if available in ADCS
     flight software (sends new calibration coefficients via telecommand).
  4. If bias is correlated with specific OBC activity, reduce or reschedule
     that activity to minimise EMI coupling.
  5. Update on-board calibration table via telecommand to compensate for
     the measured bias offset.
  6. If bias is large (>10 ?T) and persistent, switch attitude determination
     to star tracker only if available; degrade magnetorquer control.
  7. Escalate to ADCS engineering team if bias exceeds 15 ?T or does not
     respond to any corrective action within 3 orbits.


B.2 ? MAGNETOMETER SIGNAL FROZEN / STUCK VALUE
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Channels Affected: One or more of CADC0872, CADC0873, CADC0874
Anomaly Type: Type 2 (Contextual)

DESCRIPTION:
  The magnetometer channel outputs the same constant value over multiple
  telemetry samples, across orbital passes where the field should be
  continuously varying. The channel appears nominal in range but is clearly
  not sensing actual field changes.

SYMPTOM SIGNATURE IN TELEMETRY:
  - Signal variance ?² ? 0 (near-zero variance over time window)
  - Signal range = 0 (x_max - x_min ? 0)
  - Zero-crossing rate = 0
  - Autocorrelation ?? = 1.0 (perfect autocorrelation)
  - The stuck value is often the last valid reading before the fault occurred

ROOT CAUSES:
  1. ADC (Analog-to-Digital Converter) fault ? the conversion circuit has
     failed and is latching the last converted value
  2. I2C bus communication fault between sensor and OBC ? sensor is not
     responding; OBC returning last cached value
  3. Sensor firmware crash or loop hang returning a fixed value
  4. Physical sensor damage from space radiation (Total Ionising Dose effect)
  5. Power supply undervoltage preventing proper ADC operation
  6. Software bug in telemetry acquisition pipeline caching stale values

DIAGNOSTIC CHECKS:
  - Confirm the value is truly constant over > 5 minutes (rule out very
    stable orbital phase coincidence)
  - Check if other magnetometer axes are still varying ? single-axis stuck
    values strongly suggest ADC or I2C lane fault rather than field anomaly
  - Review OBC health telemetry for I2C error counters
  - Attempt to power-cycle the ADCS controller if the architecture allows
  - If the stuck value is exactly 0.0 or exactly the ADC maximum, it is
    almost certainly a hardware fault rather than a very quiet orbit

SEVERITY: HIGH ? attitude determination using magnetometers is compromised.
ADCS will compute incorrect attitude quaternion, leading to wrong control
torques. Magnetorquer-based de-tumbling will be ineffective.

CORRECTIVE ACTIONS (in priority order):
  1. IMMEDIATE: Switch attitude determination mode to exclude the stuck
     magnetometer axis if redundant sensors are available.
  2. If star tracker is available: switch primary attitude source to star
     tracker and disable magnetometer-dependent control modes.
  3. Send software reset command to ADCS magnetometer acquisition thread.
  4. Power-cycle the magnetometer sensor module (if safe to do so from
     ground telecommand).
  5. If only one of three axes is stuck: ADCS may continue with degraded
     two-axis magnetometer data. Increase Kalman filter uncertainty for
     that axis accordingly.
  6. Downgrade operational mode: switch from fine pointing to coarse
     sun-pointing while magnetometer is non-functional.
  7. If all three axes are stuck simultaneously: consider initiating safe
     mode transition to stabilise the satellite.


B.3 ? MAGNETOMETER HIGH NOISE / ELEVATED VARIANCE
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Channels Affected: One or more of CADC0872, CADC0873, CADC0874
Anomaly Type: Type 1 (Point) or Type 3 (Collective)

DESCRIPTION:
  The magnetometer signal shows rapid random fluctuations beyond the normal
  sensor noise floor, making the magnetic field vector estimate unreliable.
  The signal is moving (not stuck) but varies too erratically to represent
  the actual smoothly-varying geomagnetic field.

SYMPTOM SIGNATURE IN TELEMETRY:
  - Variance ?² significantly elevated (2-10× normal noise floor)
  - High first-difference mean (?? elevated)
  - High zero-crossing rate
  - Reduced autocorrelation at lag-1 (?? drops toward 0)
  - Signal entropy elevated

ROOT CAUSES:
  1. Active magnetorquers generating EMI ? this is the most common cause
     and is EXPECTED during de-tumble or desaturation operations
  2. EPS solar panel charge controller switching noise coupling to harness
  3. Loose electrical connection or deteriorating ground plane
  4. Radiation-induced SEU causing repeated ADC conversion errors
  5. Temperature extreme causing sensor electronics to malfunction
  6. Nearby experiment drawing pulsed power (OPS-SAT experiment payloads)
  7. Strong geomagnetic storm (Kp index > 5) amplifying natural field noise

DIAGNOSTIC CHECKS:
  - Check magnetorquer command history: if torquers were recently commanded,
    noise during torquer activity is NORMAL. Verify noise clears after
    magnetorquer deactivation (should clear within 10-30 seconds)
  - Check experiment payload activity log
  - Check EPS telemetry for battery charge/discharge events
  - Review space weather data for geomagnetic activity (Kp index)
  - Compare noise level on all three axes ? correlated noise on all three
    axes suggests a common source (power supply), uncorrelated noise
    suggests individual sensor or channel failures

SEVERITY: MEDIUM during magnetorquer activity (expected and normal).
HIGH if persistent and not correlated with known EMI sources.

CORRECTIVE ACTIONS:
  1. If correlated with magnetorquer activity: this is normal. No action.
     Allow noise to settle after torquer deactivation.
  2. Schedule magnetometer readings only during magnetorquer-off windows
     (standard operational practice, but verify it is being followed).
  3. Check and potentially halt the experiment payload generating EMI.
  4. If persistent and uncorrelated: activate ADCS noise filtering with
     increased moving-average window length.
  5. If all three axes are noisy simultaneously: consider transition to
     star tracker-only attitude determination until noise resolves.
  6. Log all noise events and correlate with EPS switching events to
     identify root cause for long-term mitigation.


B.4 ? MAGNETOMETER OUT-OF-RANGE READING
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Channels Affected: One or more of CADC0872, CADC0873, CADC0874
Anomaly Type: Type 1 (Point) or Type 2 (Contextual)

DESCRIPTION:
  The magnetometer reading exceeds the physically plausible range for
  Earth's magnetic field in LEO. Earth's field at OPS-SAT orbit altitude
  ranges from approximately 20-60 ?T depending on latitude. Readings
  outside this range indicate either sensor saturation, electronic fault,
  or significant onboard magnetic interference.

SYMPTOM SIGNATURE IN TELEMETRY:
  - Channel value > 65 ?T or < -65 ?T
  - May be a transient spike (Type 1) or persistent out-of-range condition

ROOT CAUSES:
  1. Magnetorquer near full power generating a strong local field
  2. Electromagnetic transient from EPS relay switching
  3. Sensor saturation from radiation event
  4. ADC overflow or wraparound error in software
  5. Near-polar pass where Earth's field is naturally higher (~60 ?T)

6. Software scaling error after ADCS firmware update

CORRECTIVE ACTIONS:
1. Verify orbital position ? near poles, readings up to 60 ?T are normal.
2. Check magnetorquer activity and current levels.
3. If software issue suspected: re-upload ADCS calibration tables.
4. For persistent out-of-range (>3 orbits): initiate sensor self-test.

## SECTION C: PHOTODIODE (SUN SENSOR) ANOMALY CATALOGUE

### C.1 ? PHOTODIODE DARK CHANNEL (SHADOW OR ECLIPSE CONDITION)

Channels Affected: One or more of CADC0875?CADC0880
Anomaly Type: Context-dependent ? may be NOMINAL

DESCRIPTION:
One or more photodiode channels reads near zero (dark). This is NORMAL when: (a) the channel's face is pointing away from the Sun, (b) the satellite is in eclipse, or (c) a solar panel or structure is casting a shadow. It becomes ANOMALOUS when a channel that should be seeing sunlight reads dark, or when ALL six channels read dark simultaneously outside eclipse.

DISTINGUISHING NORMAL FROM ANOMALOUS:
- Normal: One or two faces dark based on current sun vector attitude.
- Normal: All six channels dark during eclipse (lasts ~35 min per orbit).
- Anomalous: Channel remains dark when attitude suggests it should be lit.
- Anomalous: All six channels dark in expected sunlit phase of orbit.
- Anomalous: Sudden drop to dark on one channel during mid-sunlit pass.

ROOT CAUSES (Anomalous Case):
1. Photodiode failure ? sensor degraded or failed
2. Contamination ? dust or outgassing condensate on the photodiode window
3. Physical blockage by a deployed appendage or solar panel anomaly
4. Signal chain fault (ADC channel, wiring)
5. Software telemetry mapping error (wrong channel data reported)

CORRECTIVE ACTIONS:
1. Cross-reference with orbital ephemeris and attitude telemetry to determine expected Sun vector geometry ? confirm anomaly vs normal.
2. If single channel failed: note which face is affected and continue operations with remaining 5 channels for sun vector estimate.
3. If two or more channels failed: accuracy of sun vector degrades; switch to star tracker for primary attitude reference.
4. Command attitude manoeuvre to verify expected channels illuminate; this helps distinguish pointing anomaly from sensor failure.
5. If contamination suspected: enable decontamination heaters if available.

### C.2 ? PHOTODIODE SATURATION (OVERVOLTAGE)

Channels Affected: One or more of CADC0875?CADC0880
Anomaly Type: Type 1 or Type 2

DESCRIPTION:
A photodiode channel reads at or near its maximum ADC count (saturation), suggesting it is receiving more light than the sensor was designed to handle, OR the analog-to-digital conversion circuit is saturated by a

voltage exceedance.

SYMPTOM SIGNATURE:
- Channel reading at ADC maximum (e.g., 4095 for 12-bit ADC)
- Value is clipped ? no variance, stuck at maximum
- Persists across multiple telemetry samples

ROOT CAUSES:
1. Direct Solar illumination at very low Sun-angle (rare, possible near orbital plane alignment with Sun)
2. Solar concentrating reflection from Earth (albedo glint)
3. ADC reference voltage fault causing all readings to saturate
4. Short circuit in photodiode signal chain

CORRECTIVE ACTIONS:
1. Check attitude geometry ? if Sun is nearly directly face-on, saturation is physically expected and not an anomaly.
2. If not explained by geometry: verify ADC reference voltage via ADCS housekeeping telemetry.
3. If saturation on single channel: exclude from sun vector calculation and continue with remaining 5 channels.
4. Command attitude manoeuvre away from direct Sun to clear saturation.

C.3 ? PHOTODIODE NOISE / INCONSISTENT SUN VECTOR
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Channels Affected: Multiple of CADC0875?CADC0880
Anomaly Type: Type 3 (Collective)

DESCRIPTION:
Individual photodiode readings are plausible, but the sun vector computed from them is inconsistent: it may point through the satellite body, or change by more than physically possible between samples, or disagree with the expected sun position from orbital mechanics.

SYMPTOM SIGNATURE:
- Elevated variance on multiple photodiode channels simultaneously
- Computed sun vector changes direction faster than orbital dynamics allow
- Multiple channels simultaneously at intermediate voltages when only 0-1 face should be partially illuminated

ROOT CAUSES:
1. Satellite tumbling ? rapid rotation causes all photodiodes to see varying sun exposure in rapid sequence
2. Reflected sunlight from Earth or other satellite surfaces
3. Corrupted ADC sample timing ? samples from different time points are being incorrectly assembled into a simultaneous vector
4. EMI coupling from active payloads affecting analog circuits

CORRECTIVE ACTIONS:
1. Confirm satellite is not tumbling by checking ADCS rate sensor.
2. If tumbling: initiate B-dot de-tumble mode immediately.
3. Check orbital albedo prediction ? Earth-reflected light at some orbital geometries can cause inconsistent photodiode readings.
4. Increase photodiode sample averaging window to 5-10 samples.
5. If consistent and unexplained: switch primary attitude reference to star tracker.

C.4 ? PHOTODIODE GRADUAL DEGRADATION (AGING)

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Channels Affected: All of CADC0875?CADC0880 (progressive)
Anomaly Type: Type 4 (Drift)

DESCRIPTION:
  Photodiode peak voltage readings gradually decrease over weeks to months.
  This is caused by radiation damage and UV darkening of the protective
  window covering the photodiode, or by degradation of the photodiode
  junction itself.

SYMPTOM SIGNATURE:
  - Linear downward trend in peak (sunlit) voltage over multiple weeks
  - Slope of signal shows consistent downward drift
  - All six channels degrade at similar rate (if common exposure history)

ROOT CAUSES:
  1. Cumulative Total Ionising Dose (TID) from radiation belt
  2. UV darkening of optical filter or protective cover
  3. Surface contamination buildup over time

CORRECTIVE ACTIONS:
  1. Perform regular calibration using known orbital geometry with sun at
     a calibrated angle to estimate absolute sensitivity.
  2. Update photodiode gain compensation table in ADCS flight software.
  3. Monitor rate of degradation ? if accelerating, may indicate entry into
     a higher radiation environment (e.g., South Atlantic Anomaly passages).
  4. At severe degradation (>30% peak voltage loss): switch to star tracker
     as primary attitude determination for fine pointing; use photodiodes
     only for coarse sun-safe mode operations.


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END OF ANOMALY CATALOGUE ? ADCS TELEMETRY REFERENCE
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